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General

This document describes the specifications and usage of the SE-UDPS (**S**alland **E**ngineering **U**tility **D**evice **P**ower **S**upply) board. This document can be used as a guideline in conjunction with the current documentation for the Advantest V93000 SOC.

The purpose of the SE-UDPS board is to support alternative test methods for high speed interfaces (“golden device”) and overcome the limitations of physics and ATE.

Additional active circuitry on the DUT-board is used, for these active components power supplies are needed which can be controlled independently of the DUT power.

Typical usage of these power supplies will be Relay / Switches / Operational Amplifiers and buffers.

As a result the power for these components is independent of “sites”, the control of these power supplies are also independent of test sites. For DUT- circuitry, the standard supply voltages are (+5V +12V, +15V, -15V). For DC amplifiers (low voltage measurements) and I/V converters (low current measurements) thus a “clean” power is required for these devices (low noise).

The Advantest V93000 SOC can be controlled by a HP-UX as well as LINUX based workstation, the SE-UDPS is compatible on both types of system.

Specification

Configuration

Channels per board	1 each +5 V, 3 A
	1 each +12 V, 1 A
	1 each +15 V, 1 A
	1 each -15 V, 850m A

Maximum number of boards per system

Compact test head	2
Small test head	4
Large test head	8

Test head filling rules Any DPS slot can be used

Mixing The Utility Device Power Supply can integrate with other power supplies in a test head, no limitations apply.

Supply Voltage/ Current Range Specifications

Mode	Range	Max Current	Accuracy
Voltage	+ 5 V	≤3 A	±2 %
	+ 12 V	≤1 A	±2 %
	+ 15 V	≤1 A	±2 %
	– 15 V	≤850mA	±2 %

Other Performance Characteristics

General capabilities

- Over current protection
- Individual ON / OFF per power supply
- ON / OFF detection when DUT board is removed
- Kelvin connection per DPS channel (power line)

Power on settling time (within ±5%-band of nominal voltage, 47 µF external capacitive load)

- Voltage force (+5V) 2msec
- Voltage force (+12V) 6msec
- Voltage force (+15V) 3msec
- Voltage force (-15V) 3msec

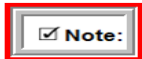
Load capacitance range

- Minimum required external capacitance of 0 µF per channel
- Maximum supported external capacitance of 470 µF per channel

Installation

The procedure of installing the SE-UDPS board is identical to each other DPS board. This installation procedure is described in Verigy V93000 SOC series HD – Service Guide Part 2, chapter 3.

After the installation procedure is completed the SE-UDPS board is ready for use, no software needs to be installed.



In order to enable the power supply the appropriate Kelvin Connection must be closed.

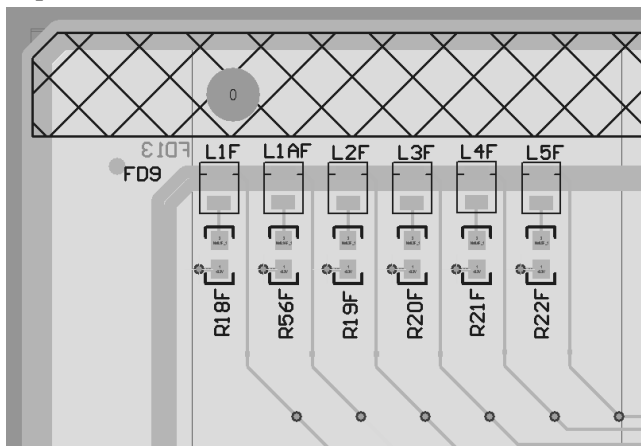
Status LED

In total 6 LED's are positioned on the SE-UDPS board which indicates the status of the board and the DPS channels. The LED's are located on the left top side.

In normal operational circumstances the LED of the corresponding DPS channel will illuminate with a green colour when the Kelvin-Point is closed.

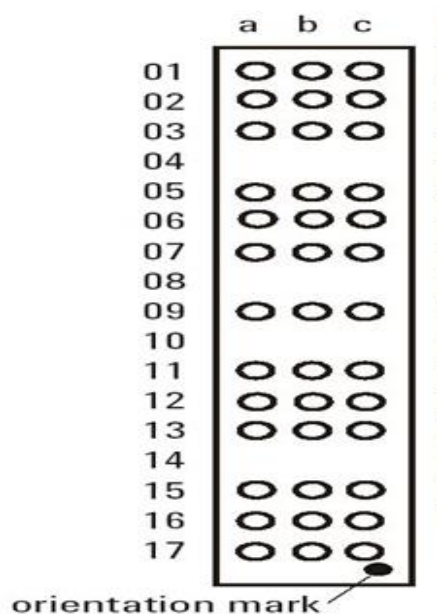
During power-on the board and online diagnostics will be executed. In case of an error a red LED will illuminate, however the corresponding green LED will not light up as a result of the Kelvin-Point being closed.

L1F (red)	→	Indicates error situation
L1AF (yellow)	→	Indicates warning (Future purpose)
L2F (green)	→	DPS channel 4
L3F (green)	→	DPS channel 3
L4F (green)	→	DPS channel 2
L5F (green)	→	DPS channel 1



Pinning Pogo cable

The SE-UDPS uses the existing GP-DPS / MS-DPS pogo cable, the picture below displays the pinning of the original cable.



Pin	a	b	c
1	GND	NC	NC
2	GND	NC	Sense-1
3	NC	NC	Force-1
4	NC	NC	NC
5	GND	NC	NC
6	GND	NC	Sense-2
7	NC	NC	Force-2
8	NC	NC	NC
9	NC	NC	NC
10	NC	NC	NC
11	GND	NC	NC
12	GND	NC	Sense-3
13	NC	NC	Force-3
14	NC	NC	NC
15	GND	NC	NC
16	GND	NC	Sense-4
17	NC	NC	Force-4

The SE-UDPS uses no 'Guard' and 'Ground sense' pins, this results in the following pinning for the SE-UDPS, see the table below.

Channel 1 → P5V / 3A
Channel 2 → P12V / 1A
Channel 3 → P15V / 1A
Channel 4 → M15V / 1A



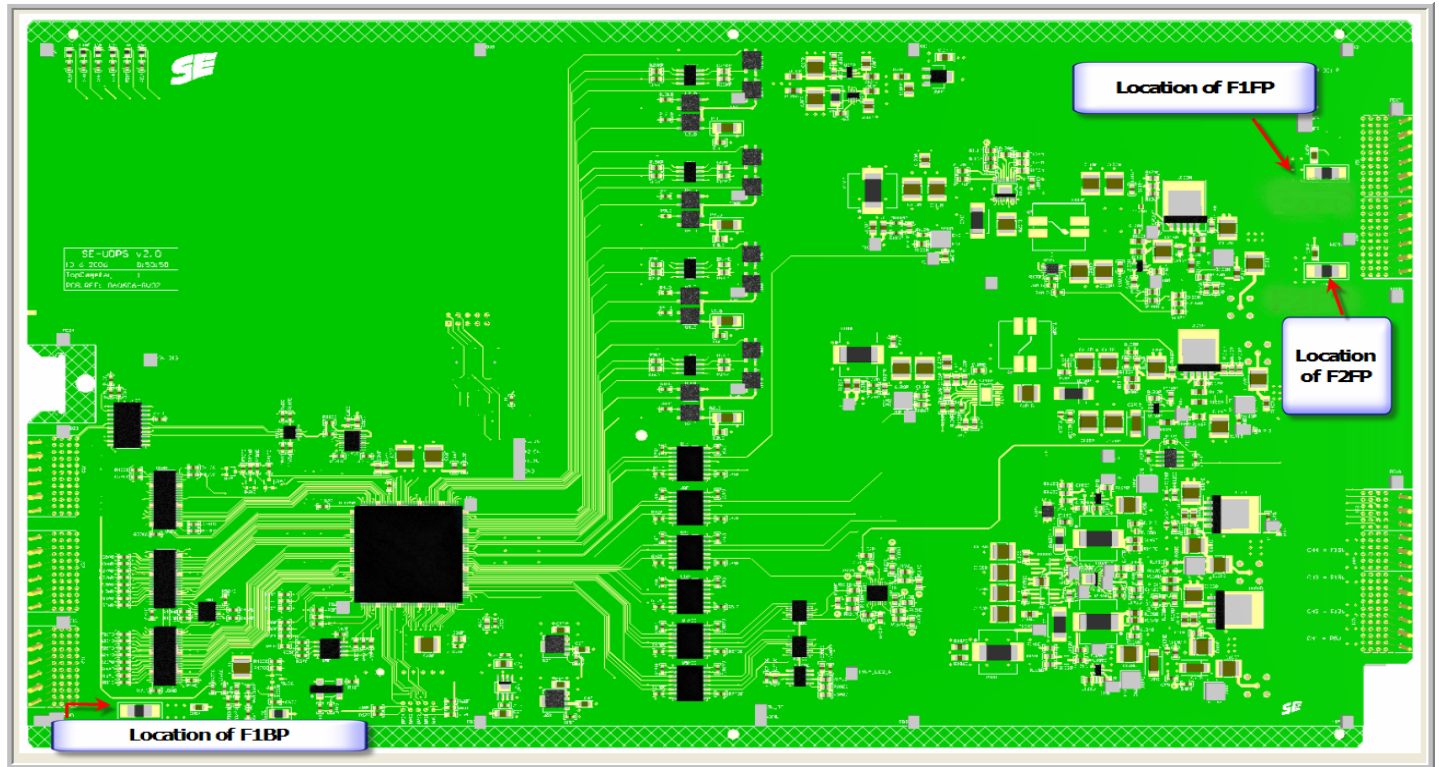
Trouble shooting

This chapter describes what to do if you encounter a problem with the SE-UDPS module for the Verigy V93000 SOC Series.

Failure	Solution....
The test system cannot be started after the installation of SE-UDPS	• Check / copy service_guide_part_I_E6978-91004_2.7.pdf , page 34
Red LED illuminates	• The on-board diagnostics has detected an error during start-up of the board. Check if there is no short circuit on the load-board and re-start the test system.
No green LED illuminates	• This indicates that no power supply is enabled; verify if the appropriate Kelvin Point has been closed.
Green LED is blinking	• A blinking LED indicates that the load is too heavy to enable the power supply.
No voltage on one or more DPS channels after power-up test system	• Check if there is no red LED lighting up on the SE-UDPS board. If the red LED is illuminated, the on-board diagnostics has detected an error during start-up of the board. Check if there is no short-circuit on the load-board and re-boot. • Check if there is a green LED blinking on the SE-UDPS board. A blinking LED indicates that the load is too heavy to enable the DPS channel. • Check if the Kelvin Point (KP) has been closed of the corresponding DPS channel. If not, please close the KP for that channel, the green LED must lighting up and voltage will be present on the pins. • Remove the board and check if all the fuses are still ok (F1BP / F1FP / F2FP) for a visual overview of the fuse locations see SE-UDPS fuse locations



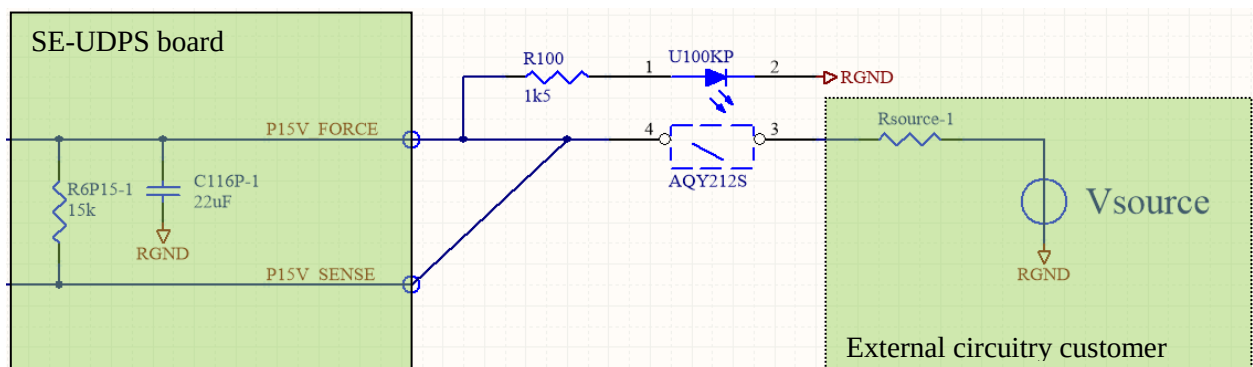
SE-UDPS fuse locations



Application note – External stimuli during docking

An external voltage applied to the DPS-channel during docking of the load board can impact the internal mechanism to detect the Kelvin-Point and prevent the DPS-channel from turning on this DPS-channel. In case the impedance of the external voltage source is high ($> 1k\Omega$) than this issue will not occur. With a lower impedance Salland Engineering recommends adding a photomos-relay to the DPS signal path on the load board. Depending on the load-voltage, load-current and acceptable voltage-drop a wide-range of photomos devices is available.

An example is given below, including schematic.



Panasonic

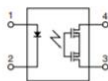
Automation Controls Catalog

Miniature SOP4-pin type
with high capacity
up to 1.6A

PhotoMOS[®]
GU SOP 1 Form A High Capacity
(AQY210GS)



mm inch



RoHS compliant

FEATURES

1. Continuous load current: Max. 1.6A high capacity (AQY211G2S)
2. Low on resistance: typical 0.1 Ω (AQY211G2S)
3. Broad lineup of high capacity types

TYPICAL APPLICATIONS

- Measuring instruments
- Security and disaster-preventing system: use in I/O for alarm and security devices, etc.

TYPES

	Output rating*	Package	Part No.			Packing quantity	
			Tube packing style	Tape and reel packing style		Tube	Tape and reel
				Picked from the 1/2-pin side	Picked from the 3/4-pin side		
AC/DC dual use	40V 1.6A 1.25A 60V 1.0A	SOP4-pin	AQY211G2S	AQY211G2SX	AQY211G2SZ	1 tube contains: 100 pcs. 1 batch contains: 2,000 pcs.	1,000 pcs.
			AQY212G2S	AQY212G2SX	AQY212G2SZ		
			AQY212GS	AQY212GSX	AQY212GSZ		

* Indicate the peak AC and DC values.

Note: For space reasons, the three initial letters of the part number "AQY", the surface mount terminal shape indicator "S" and the packing style indicator "X" or "Z" are not marked on the device. (Ex. the label for product number AQY212G2SX is 212G2.)

(Available at e.g. Digikey: 255-3566-1-ND)